

WILLOW CREEK, AB, Canada

WMO: 712520

Lat: 53.386N

Lon: 118.347W

Elev: 4495

StdP: 12.46

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.2	-22.8	-36.8	0.8	-29.9	-28.6	1.3	-22.2	10.0	20.2	8.6	18.9	0.5	0	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.0	79.1	54.9	75.3	53.7	71.8	52.4	57.3	72.3	55.6	70.5	54.0	68.1	5.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.4	69.2	58.6	50.4	64.1	57.3	48.5	59.5	55.9	26.5	72.2	25.3	70.5	24.3	68.7	64.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	7.7	-38.3	85.3	3.1	2.9	-40.5	87.4	-42.3	89.1	-44.1	90.8	-46.4	92.9		
(5)			-38.2	61.2	2.8	1.6	-40.2	62.4	-41.9	63.3	-43.5	64.2	-45.6	65.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.6	13.6	15.2	22.6	32.8	42.0	48.6	53.4	52.2	44.7	34.4	20.8	10.2		
	DBStd	17.95	14.29	13.28	12.37	7.83	6.45	5.07	5.37	5.37	6.66	7.97	12.33	13.17		
	HDD50	6653	1128	975	849	515	258	84	27	40	184	484	877	1232		
	HDD65	11811	1593	1395	1314	965	714	492	360	396	610	949	1326	1697		
	CDD50	317	0	0	0	0	8	43	132	109	24	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	449	0	0	0	0	6	40	216	146	41	0	0	0		
	CDH80	67	0	0	0	0	0	6	36	22	3	0	0	0		
	WSAvg	2.6	2.2	2.5	2.8	3.1	3.2	2.9	2.8	2.6	2.5	2.5	2.2	1.8		
	PrecAvg	25.3	1.7	1.1	1.5	1.3	2.3	3.5	3.5	2.8	2.0	1.9	2.0	1.7		
(15) Precipitation	PrecMax	31.7	3.4	2.1	2.5	2.5	4.3	5.6	7.0	4.3	3.9	3.1	4.0	2.9		
	PrecMin	19.4	0.4	0.4	0.4	0.6	1.2	1.1	1.7	1.4	0.7	1.0	0.7	0.8		
	PrecStd	2.7	0.7	0.4	0.5	0.5	0.7	1.1	1.3	0.8	0.7	0.6	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	46.9	55.8	65.4	74.4	79.5	84.4	83.0	79.0	67.2	52.4	42.3		
		MCWB	36.0	35.7	40.4	45.1	50.1	54.1	56.1	56.5	53.9	48.1	41.6	34.7		
	2%	DB	41.0	41.6	48.5	58.4	69.5	73.1	79.7	78.3	73.7	59.1	44.6	37.2		
		MCWB	33.8	32.5	36.6	42.3	48.1	52.1	55.0	55.6	52.5	45.2	37.3	32.0		
	5%	DB	36.9	37.9	44.0	52.7	64.5	69.0	75.7	74.0	68.5	53.0	39.6	32.6		
		MCWB	31.2	30.4	34.0	39.1	47.3	50.8	54.2	54.2	50.7	42.3	34.2	29.2		
	10%	DB	32.8	34.1	40.3	48.3	59.8	64.8	71.3	69.8	61.7	47.8	35.5	28.8		
		MCWB	28.9	28.3	31.7	37.1	44.9	48.9	53.1	53.1	48.3	39.2	31.8	26.3		
	0.4%	WB	37.1	36.7	40.7	46.0	52.3	56.7	59.4	60.0	56.1	49.1	42.1	35.9		
		MCDB	43.0	44.8	54.4	63.3	66.3	70.3	74.2	72.6	72.4	64.2	49.8	41.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.1	33.2	37.4	43.0	49.9	54.3	57.1	57.7	53.5	45.9	38.0	32.4		
		MCDB	40.0	40.2	46.4	56.0	64.9	68.4	72.2	72.2	70.3	56.3	43.7	36.2		
	5%	WB	31.8	31.1	34.7	40.2	47.8	52.1	55.4	55.8	51.5	43.1	34.9	29.3		
		MCDB	36.6	37.1	42.7	51.1	62.0	65.2	71.5	69.9	66.5	52.0	38.7	32.4		
	10%	WB	28.9	28.6	32.5	37.5	45.8	50.2	54.0	54.0	49.0	40.2	32.3	26.3		
		MCDB	32.6	33.8	39.2	47.2	58.6	62.2	68.5	67.1	60.9	46.6	34.6	28.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.9	28.6	27.9	24.8	26.3	26.8	29.0	29.8	28.1	23.7	21.8	23.8		
		MCDBR	23.5	28.6	30.0	33.3	38.4	37.1	40.9	40.9	43.4	32.8	22.4	21.5		
		MCWBR	16.8	20.3	19.1	18.4	19.6	18.0	18.7	20.2	22.7	19.3	15.7	16.7		
		MCDBR	21.8	27.1	27.9	30.4	34.4	31.7	34.3	35.3	38.8	30.2	20.5	21.1		
	5% WB	MCWBR	15.9	19.3	18.0	17.2	18.3	16.1	16.6	18.3	20.9	18.3	14.9	16.8		
(40) Clear-Sky Solar Irradiance	taub		0.240	0.245	0.280	0.310	0.331	0.346	0.336	0.337	0.303	0.278	0.246	0.232		
	taud		2.431	2.486	2.391	2.333	2.385	2.392	2.462	2.448	2.553	2.574	2.481	2.364		
	Ebn at Noon		248	285	291	293	291	286	288	281	281	267	244	226		
	Edh at Noon		16	21	29	35	35	36	32	31	25	19	15	14		
(44) All-Sky Solar Radiation	RadAvg		249	520	866	1336	1566	1617	1681	1432	996	555	267	194		
	RadStd		22	53	72	102	139	136	140	92	117	47	27	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page